Name_____ Date_____ Period_____

LESSON 5-7 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Surface Area of Rectangular Prisms

Maryland Standard 6.G.A.4 · Learning target – I can find the surface area of a rectangular prism by adding the areas of its six faces.

REMEMBER

- **Surface area** is the total area of all the faces, in **square units**.
- A rectangular prism has 3 pairs of matching faces: **SA = 2lw + 2lh + 2wh**.

WORKED EXAMPLE

A box is 8 in × 3 in × 2 in. Find its surface area.

Step 1 One face from each pair: 8 × 3 = 24, 8 × 2 = 16, 3 × 2 = 6

Step 2 Add: 24 + 16 + 6 = 46

Step 3 Each face has a twin, so double: 2 × 46 = 92

Answer **92 in²**

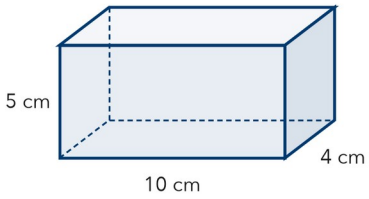
WORD BANK

surface area – the total area of all the faces
face – a flat side of a solid
pair – two matching faces
square units – cm², in², ft²

PRACTICE

show your work · label units

1. Find the surface area of the prism.



Step 1 Top and bottom: 2 × (10 × 4).

Step 2 Front and back: 2 × (10 × 5).

Step 3 Two ends: 2 × (4 × 5). Then add.

Show your work.

Answer_____ cm²

2. Find the surface area of each solid.

Step 1 Cube: 6 × (edge × edge).

Step 2 Prism: 2lw + 2lh + 2wh.

a. cube, edges 3 cm _____ cm²

b. cube, edges 10 ft _____ ft²

c. prism 4 m × 2 m × 1 m _____ m²

d. prism 5 in × 5 in × 2 in _____ in²

e. cube, edges 5 in _____ in²

f. prism 6 cm × 3 cm × 2 cm _____ cm²

3. Gift Box. Nadia wraps a box. It is 12 in long, 8 in wide, and 4 in tall. Her paper covers 400 in². Is that **enough** paper? Explain.

Step 1 Find the surface area of the box.

Step 2 Compare it to 400 in².

Show your work.

Answer_____

4. Toy Chest. A toy chest is 4 ft long, 2 ft wide, and 3 ft tall. Leo paints all faces but the bottom. How many ft² does he paint?

Step 1 Find the top only (no bottom).

Step 2 Find the front and back, then the two sides.

Step 3 Add the 5 faces.

Answer_____ ft²

5. Find the Error. Diego says the surface area of a cube with 2 cm edges is 8 cm² because 2 × 2 × 2 = 8. Explain his mistake and find the surface area.

Frame *Diego found the ___, not the surface area. A cube has ___ faces, so ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Pick a box at home. Point to each pair of matching faces and tell someone how you would find its surface area.

Name_____ Date_____ Period_____

LESSON 5-7 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Surface Area of Triangular Prisms

Maryland Standard 6.G.A.4 · Learning target – I can find the surface area of a triangular prism using its net.

REMEMBER

- A triangular prism has **2 triangle bases** and **3 rectangles**.
- Each rectangle is one side of the triangle × the prism's length. Add all **5 faces**.

WORKED EXAMPLE

Prism 10 cm long. Ends are right triangles with sides 3, 4, 5 cm.

Step 1 Triangles: 2 × (½ × 3 × 4) = 12

Step 2 Rectangles: 30 + 40 + 50 = 120

Step 3 Add: 12 + 120 = 132

Answer **132 cm²**

WORD BANK

triangular prism – 2 triangle ends + 3 rectangles

rectangle face – triangle side × prism length

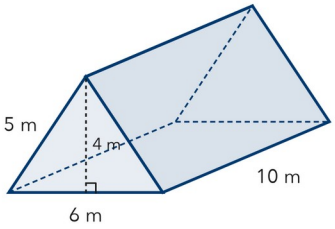
height – straight up from the base, at a square corner

surface area – add all 5 faces

PRACTICE

show your work · label units

1. Find the surface area. The triangle ends have sides 5 m, 5 m, and 6 m.



Step 1 Two triangles: 2 × (½ × 6 × 4).

Step 2 Rectangles: each side × 10.

Step 3 Add all 5 faces.

Show your work.

Answer_____ m²

2. Find each area.

a. 2 triangles, base 8 ft, height 3 ft: _____ ft²

b. 1 rectangle, 5 ft by 9 ft: _____ ft²

3. Tent. A tent is a **triangular prism**, 9 ft long. Each end is a triangle: base 8 ft, height 3 ft, and two 5 ft sides. Find the fabric for **all 5 faces**, with the floor.

Step 1 Find the two ends.

Step 2 Find the floor and the two sides.

Step 3 Add.

Answer_____ ft²

4. Doorstop. A wooden doorstop is a triangular prism 4 in long. Its ends are right triangles with sides 5 in, 12 in, and 13 in. Find its surface area.

Step 1 Ends: the legs 5 in and 12 in are the base and height.

Step 2 Rectangles: 5 × 4, 12 × 4, 13 × 4.

Step 3 Add all 5 faces.

Answer_____ in²

5. Find the Error. For the prism in Exercise 1, Lucas found the rectangles as 6 × 10, 4 × 10, and 4 × 10. Explain his mistake.

Frame *Lucas used the triangle's ___ instead of its ___ for two rectangles.*

_____

_____

TEST PRACTICE

choose the best answer

6. Triangle ends: 9 cm² each. Rectangles: 20, 20, and 24 cm². What is the surface area?

A 64 cm² **B** 73 cm² **C** 82 cm² **D** 146 cm²

7. A prism is 8 cm long. Its triangle ends have sides 6, 8, and 10 cm. What is the total area of the 3 rectangles?

A 24 cm² **B** 192 cm² **C** 240 cm² **D** 480 cm²

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___